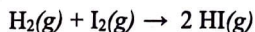


Unit 4 Practice MCQ Questions

1. The equilibrium constant, K_c , for the reaction



is 55 at 425°C. The value for the equilibrium constant will be ~~changed if~~ *increased*

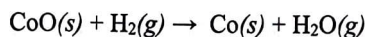
$$\Delta H = -100 \text{ kJ/mol}$$

- A. ~~concentrations are given in atmospheres instead of mol/L. $T = 500 \text{ K}$~~
 B. a catalyst is added to the reaction system.
 C. the size of a reaction vessel is doubled.
☒ D. the temperature is changed to 400°C.
 E. additional HI is added to the reaction system.

2. Which of the following statements is true in a reaction system at equilibrium?

- A. The number of collisions per unit time between reactants is equal to the number of collisions per unit time between products.
☒ B. Reactants are reacting to form products at the same rate as products are reacting to form reactants.
 C. The product of the concentrations of the reactants is equal to the product of the concentrations of the products.
 D. All concentrations of reactants and products are the same.
 E. No products are being formed, and no reactants are being formed.

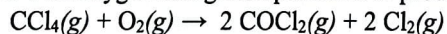
3. For the reaction system



at 550°C, $K = 67$. The equilibrium constant expression is

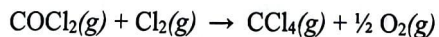
- A. $\frac{[\text{CoO}][\text{H}_2]}{[\text{Co}][\text{H}_2\text{O}]}$
 B. $\frac{[\text{Co}][\text{H}_2\text{O}]}{[\text{CoO}][\text{H}_2]}$
 C. $\frac{[\text{Co}][\text{H}_2\text{O}]}{[\text{H}_2]}$
 D. $\frac{[\text{H}_2]}{[\text{H}_2\text{O}]}$
☒ E. $\frac{[\text{H}_2\text{O}]}{[\text{H}_2]}$

4. Carbon tetrachloride reacts with oxygen at high temperatures to produce chlorine and carbonyl chloride.2



$$K_c = 1.9 \times 10^{19}$$

What is K_c for the following:



← reversed $\pm$ $\frac{1}{2}$ original reaction!

- A. 5.2×10^{-20}
☒ B. 2.3×10^{-10}
 C. 4.4×10^9
 D. -1.9×10^{19}
 E. 1.9×10^{19}

$$K' = \frac{1}{K^{1/2}} = \frac{1}{(1.9 \times 10^{19})^{1/2}}$$

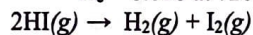
5. Consider the equilibrium $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$ at a certain temperature. An equilibrium mixture in a 4.00-L vessel contains 0.40 M NH_3 , 0.200 M N_2 , and 0.30 M H_2 . What is the value of K_c ?

- A. 9.00
☒ B. 29.6
 C. 3.37
 D. 17.1
 E. 7.41

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{(0.40)^2}{(0.200)(0.30)^3} = 29.6$$

6. For the reaction system:

$$K_c = 0.020 \text{ at } 720 \text{ K.}$$



If the initial concentrations of HI, H₂, and I₂ are all $1.50 \times 10^{-3} \text{ M}$ at 720 K, which one of the following statements is correct?

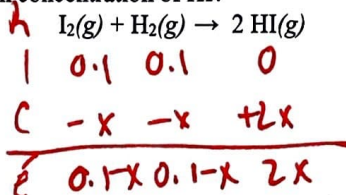
- A. The system is at equilibrium. $Q \neq K$
 B. The concentrations of HI and I₂ will increase as the system is approaching equilibrium.
 C. The concentrations of H₂ and HI will decrease as the system is approaching equilibrium.
 D. The concentration HI will increase as the system is approaching equilibrium.
 E. The concentrations of H₂ and I₂ will increase as the system is approaching equilibrium.

$$Q = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} = \frac{1.5 \times 10^{-3} \cdot 1.5 \times 10^{-3}}{(1.5 \times 10^{-3})^2} = 1 > K$$

$Q > K$ reverse!

7. In an experiment, 0.100 mol H₂ and 0.100 mol I₂ are mixed in a 1.00-L container and the reaction forms HI. If $K_c = 50.0$ for this reaction, what is the equilibrium concentration of HI?

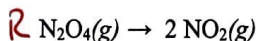
- A. 0.156 M = 0.0779×2
 B. 0.0780 M
 C. 7.07 M
 D. 0.200 M
 E. 0.0384 M



$$K = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = 50.0 = \frac{(2x)^2}{(0.1-x)^2}$$

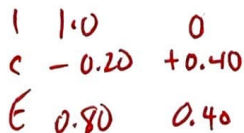
$$x = 0.0779$$

8. Exactly 1.0 mol N₂O₄ is placed in an empty 1.0-L container and is allowed to reach equilibrium described by the equation



If at equilibrium the N₂O₄ has a concentration of 0.80 M, what is the value of the equilibrium constant for the reaction under these conditions?

- A. 0.05
 B. 0.2
 C. 0.5
 D. 20
 E. 400



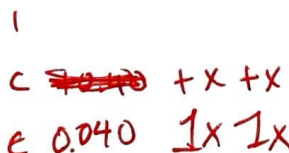
$$K = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{0.40^2}{0.80} = 0.2$$

9. For the equilibrium



$K_c = 4.0$ at 228°C. Pure PCl₅ is added to the reaction system. At equilibrium there is 0.040 M PCl₅. What is the concentration of PCl₃ in this system?

- A. 0.16 M
 B. 0.20 M
 C. 0.40 M
 D. 0.80 M
 E. 1.6 M

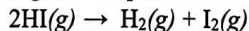


$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = 4$$

$$\frac{x^2}{0.040} = 4$$

$$x^2 = 0.16 \quad x = 0.4$$

10. Hydrogen iodide undergoes decomposition according to the equation



The equilibrium constant at 425°C for this system is 0.018. If 1.0 mol each of H₂, I₂, and HI were placed together in a 1-L container at 425°C, then

- A. the concentration of HI would decrease. ✓
 B. the value of K would increase to 1.0. X
 C. because of reaction, the total number of molecules would decrease. X
 D. because of reaction, the total number of molecules would increase. X
 E. the concentration of H₂ would decrease. X

11. Which of the following equilibria would not be affected by pressure changes at constant temperature?

- A. FeO(s) + CO(g) → Fe(s) + CO₂(g) NO 1:1 gas!
 B. CaCO₃(s) → CaO(s) + CO₂(g) Yes 0:1 moles
 C. H₂O(g) → H₂(g) + ½ O₂(g) Yes 1:1.5 moles
 D. 2 NO(g) + O₂(g) → 2 NO₂(g) Yes 3:2 moles
 E. PCl₅(l) → PCl₃(g) + Cl₂(g) Yes 0:2 moles

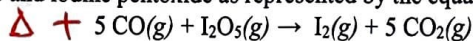
12. Consider the reaction



If the above reactants and products are contained in a closed vessel and the reaction system is at equilibrium, the number of moles of CS_2 can be increased by

- A. adding some S_2Cl_2 to the system. ~~x~~ (liquid)
 B. removing some CCl_4 from the system. ~~x~~ (liquid)
 C. decreasing the size of the reaction vessel. ~~No~~ $\rightarrow$ ~~more~~ CS_2
 D. increasing the temperature of the reaction system. $\checkmark$
 E. adding some Cl_2 to the system. ~~No~~ $\rightarrow$ reverse shift

13. The reaction of carbon monoxide and iodine pentoxide as represented by the equation



is endothermic. The yield could be increased by

yield = production or # products

- A. increasing the pressure. ~~x~~ 6:6 moles gas
 B. decreasing the pressure. ~~x~~ 6:6 moles gas
 C. increasing the temperature. $\rightarrow$
 D. decreasing the temperature. ~~Nope~~
 E. decreasing the volume of the reaction vessel. ~~No~~ 6:6 ratio

14. Which of the following equilibria would not be affected by volume changes at constant temperature?

- A. $2 \text{CO}(g) + \text{O}_2(g) \rightarrow 2 \text{CO}_2(g)$ 3:2 ~~No~~
 B. $2 \text{NO}_2(g) \rightarrow \text{N}_2\text{O}_4(g)$ 2:1 ~~No~~
 C. $2 \text{NO}(g) + 3 \text{F}_2(g) \rightarrow 2 \text{F}_3\text{NO}(g)$ 5:2 ~~No~~
 D. $\text{BF}_3(g) + \text{NH}_3(g) \rightarrow \text{H}_3\text{NBF}_3(s)$ 2:0 ~~No~~
 E. $\text{O}_3(g) + \text{NO}(g) \rightarrow \text{NO}_2(g) + \text{O}_2(g)$ 2:2 $\checkmark$ Yes!

15. For the reaction system $\text{H}_2(g) + \text{I}_2(g) \rightarrow 2 \text{HI}(g)$, $K = 55$ at 425°C .

For the system $\text{HI}(g) \rightarrow \frac{1}{2} \text{H}_2(g) + \frac{1}{2} \text{I}_2(g)$, K at 425°C is

- A. -55
 B. 55
 C. 0.13
 D. 0.018
 E. 3.3×10^{-4}

reverse rxn, $1/2$ rxn

$$K' = 1/K^{1/2} = 1/(55)^{1/2}$$

16. For which of the following systems at equilibrium and at constant temperature will decreasing the volume cause the equilibrium to shift to the right?

- A. $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$ 4 $\rightarrow$ 2 $\checkmark$
 B. $2 \text{H}_2\text{O}(g) \rightarrow 2 \text{H}_2(g) + \text{O}_2(g)$ 2 $\rightarrow$ 3 ~~x~~
 C. $2 \text{NO}_2(g) \rightarrow 2 \text{NO}(g) + \text{O}_2(g)$ 2 $\rightarrow$ 3 ~~x~~
 D. $\text{NH}_4\text{Cl}(s) \rightarrow \text{NH}_3(g) + \text{HCl}(g)$ 0 $\rightarrow$ 2 ~~x~~
 E. $\text{H}_2(g) + \text{Cl}_2(g) \rightarrow 2 \text{HCl}(g)$ 2 $\rightarrow$ 2 ~~x~~

toward less moles of gas

17. In which of the following reactions does an instantaneous increase in the volume of the reaction vessel favor formation of the products?

- A. $\text{MgO}(s) + \text{CO}_2(g) \rightarrow \text{MgCO}_3(s)$ 1 $\rightarrow$ 0 ~~No~~
 B. $\text{PCl}_5(g) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$ 1 $\rightarrow$ 2 $\checkmark$ Yes!
 C. $\text{H}_2(g) + \text{I}_2(g) \rightarrow 2 \text{HI}(g)$ 2 $\rightarrow$ 2 ~~No~~
 D. $\text{N}_2(g) + \text{O}_2(g) \rightarrow 2 \text{NO}(g)$ 2 $\rightarrow$ 2 ~~No~~
 E. $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$ 4 $\rightarrow$ 2 ~~No~~

$\rightarrow$ toward more moles of gas

18. Consider the following reaction. $2 \text{CO}(g) + \text{O}_2(g) \rightarrow 2 \text{CO}_2(g)$

If equimolar amounts of CO and O_2 gas are introduced into the system and the system is allowed to come to equilibrium, which statement is correct?

- A. $P_{\text{CO}} < P_{\text{O}_2} < P_{\text{CO}_2}$
 B. $P_{\text{CO}} = P_{\text{O}_2} < P_{\text{CO}_2}$
 C. $P_{\text{CO}_2} < P_{\text{CO}} < P_{\text{O}_2}$
 D. $P_{\text{CO}_2} < P_{\text{CO}} = P_{\text{O}_2}$

$$K_p = 2.9 \times 10^3$$

$\rightarrow$ CO consumed @ $2 \times \text{O}_2$!
 $\text{CO} < \text{O}_2$

$\rightarrow$ Products will be greater than reactants b/c $K > 1$